Non-LLM Agent Ecosystem — Master Design

All 3 classes · structure · architecture · training · the metrics bible · research-paper map · gates | v1 · 2026-10-26 · **GFI planning corpus** · system name pending owner reveal

The one-skeleton, three-species picture. Every agent of every class shares identical anatomy — only the action space, the reward, and the auxiliary math differ. Nothing on the hot path is an LLM.

```
 engines' data (PIT, Arrow) ⇒ SHARED ANATOMY:
   feature bus (µs, mmap) → typed head {action, size, conf(calibrated), ttl}
   → deterministic risk gateway (rules, never learned) → unified order model → venue
 offline: PIT-replay train → walk-forward+CPCV eval → shadow → HUMAN gate → deploy → monitor → nightly retrain ⇄
rollback(signed)
   TYPE 1 alpha cells (direction edge)   TYPE 2 market makers (spread+inventory edge)   TYPE 3 portfolio layer
(allocation edge)
```

Honest capability: internal loop µs-class (CI-benchmarked); venue hop 0.5–50 ms external; targets intraday→multi-day edges + 24/7-venue MM — not sub-ms colo HFT (R8, upgrade path preserved).

1 · Shared anatomy & the Head Contract

Stage	Piece	Spec (class-independent)
Ingest	typed events	envelope: event_id, schema_v, source_id, occurred_at/ingested_at, rights_class, lineage, hazard_flags
Feature	mmap Arrow columns	0.1–2 µs read; semver + lineage; training reads THE SAME bytes as inference
Head	policy payload	{action_typed, size, conf: calibrated, ttl}
Gateway	deterministic rules	position caps · loss ladders · kill-switch · market-state gates — never learned (R7)
Encode	unified order model	same encoding for every venue; sim↔live bit-identical
Observe	attribution logger	PnL decomposed per trade: why made/lost (features + regime + execution cost)

The Head Contract (Jev/Laya-pattern — verified as an extern class; owned natively at µs):

```
Type 1: {target_position_delta | spread_delta | greek_targets, urgency, conf, ttl}
Type 2: {bid_price, bid_size, ask_price, ask_size, skew, conf, ttl}
Type 3: {target_weights[], order_intents, risk_budget}
```

Confidence = **conformal-calibrated** on a frozen window (P(profitable|c) measured); miscalibration ⇒ auto-demote. Confidence mechanically drives **position sizing · peer-trust weights · allocator belief-quality**. TTL kills stale decisions. No text, no generation, one forward pass, weights resident — non-LLM by structure, not by policy promise.

Lifecycle gates (all classes)

```
train(PIT replay) → evaluate(walk-forward+CPCV,
champion race)
→ certify(PSR/DSR) → shadow(2–4wk live, zero risk)
→ approve(HUMAN, graduated capital) → deploy
→ monitor(attribution+drift+SLO) → nightly retrain ⇄
rollback(signed, seconds)
```

Agents never touch own risk limits; one live champion per unit; candidates live in shadow rings.

Class-appropriate champion controls — every candidate must beat its deterministic control:

Class	Must beat
T1 alpha	GBM/linear forecast-and-trade
T2 MM	static Avellaneda-Stoikov core
T3 portfolio	inverse-vol static allocator

2 · TYPE 1 Alpha cells — quant · stat-arb · derivatives · event-driven

Cell unit: one cell = one tradeable decision — a symbol, OR a PAIR/basket for stat-arb (RELIANCE-INFY spread · BTC-ETH ratio): the trade IS the relative value, so the cell's world-model is the spread, not the legs.

Variant	Action space	Auxiliary learned models	Reward
Intraday / mid-freq	target Δ position ±K lots, or cancel	— (policy is the product)	ΔPnL − fees − slippage − λ·turnover
Stat-arb pair/basket	joint leg Δ	cointegration-strength · spread drift (nightly re-est)	spread PnL + alignment penalty
Futures	target Δ	basis/rollover calendar · term-structure state	PnL − rollover penalty
Options	greek targets (Δ, Vega, θ-budget)	vol-surface learner; **Greeks engine stays deterministic**	PnL − γ-scalping friction
Event-driven	Δ + urgency	event-response curves per event-type (News/World engines + event-study priors); corroboration-weighted signals	event-window PnL − false-positive cost

Model & training program · **family trunks, symbol heads** (refined 2026-10-26): one trunk PER ASSET-CLASS FAMILY (equities · crypto · futures… — microstructure genuinely differs across classes), symbol personality learned in small heads; strategy differences emerge from head + per-cell evolved risk params + regime heads + per-event-response curves — never one trunk imposing one strategy (QuantNet, QF 2022: family-agnostic features + market-specific heads beat both extremes) · shared-trunk encoder (learned from ALL symbols of the family) + small per-cell head — new symbol = instant competence, then specialize · algo ladder: ① supervised baseline (permanent champion) → ② RL fine-tune: SAC (continuous sizing) / PPO (act·cancel) → ③ pycma outer evolution (hyperparams, risk params, feature sets) · **regime dual-heads**: one head per regime bucket (calm · volatile · event-window); detector selects; certification is per-head · nightly: replay simulates years; incremental fine-tune + periodic full retrain; seed-invariance CI (same seed ⇒ same weight hash)

What matters most (expert order) 1. **expectancy AFTER costs** — a 2 bp gross signal paying 2.5 bp is noise; turnover drag always in reward 2. impact/slip modeling before scaling — capacity curves, participation caps 3. tails: Sharpe is tail-blind — short-vol trap; CVaR in the risk budget 4. non-stationarity as a first-class OBSERVATION (regime in obs space) 5. event-cells: false positives COSTED, never free

3 · TYPE 2 Market-making — quoting · crypto MM · spread capture

Species difference: the product is spread + inventory control, not direction. Highest latency-sensitivity of all classes; adverse selection is the permanent enemy.

```
fair value (learned: mid + short-horizon drift)
  → Avellaneda-Stoikov-class quote core (deterministic spread+skew math, provable)
  → 2 small learned nets: size adj · skew adj (within certified bounds — never end-to-end RL quoting)
  → quote risk grid (deterministic: min/max spread, max size, inventory cap)
  → dialer (core-pinned thread, per-event re-price, tens of µs)
```

Reward = spread_capture − adverse_selection − inventory_risk − fees(rebates) · inventory = first-class: governor **auto-widens** as inventory nears cap · one-sided pausing under uncertainty. **Survival gear:** poison-tape detector (pull quotes on stale/crossed/flash) · quote-storm breaker · **front-run firewall: MM never front-runs the portfolio's own parent orders**. **Entry:** crypto venues first; MM's realized fills calibrate the per-venue cost model ALL classes consume.

4 · TYPE 3 Portfolio & asset-management layer

Species difference: money-management errors are the costliest errors ⇒ deterministic optimization cores with LEARNED INPUTS, never a big black box.

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learned inputs                                      deterministic cores
belief signals (cells: signed, confidence)  →  allocation engine (Black-Litterman-class posterior blend)
regime states (detectors, dwell-confirmed)  →  risk targeter (σ-target vol scaling + DD de-risk ladder)
covariance (Ledoit-Wolf shrink, regime-cond) →  rebalancer (λ·turnover cost-aware, band-triggered)
cost model (per venue, MM-calibrated)       →  governor overlay (caps, loss ladders — human owns limits)
```

Allocator = the ONLY mover of capital ACROSS cells; hysteresis bands (no churn from micro-signals) · σ walks to target via ladder; DD states map to de-risk steps — a **deterministic table, not learned** (R7). **What matters most:** 1. estimation risk > optimizer choice (wrong Σ poisons any allocator) · 2. turnover drag gates every rebalance · 3. cross-agent crowding = ONE correlated bet → systemic circuit breaker · 4. dwell-time regime confirmation before allocation changes.

5 · THE METRICS BIBLE — what matters from a top-firm quant's chair

5.1 EDGE family — is there money in the idea?

Metric	Definition	Honest role
Expectancy	$E = p \cdot \bar{W} - (1-p) \cdot \bar{L}$ (+costs!)	THE first-class edge number; everything else is a view of it
Win rate	fraction profitable	diagnostic only — meaningless alone
R:Reward	$\bar{W} / \bar{L}$ (R-multiples)	the other half of the pair
Break-even WR	$p^* = 1/(1+R:R)$ + frictions	sanity floor
Profit factor	Σwins/Σlosses	scale-robust expectancy (v1 target >1.3 MF-cells)

Win-rate × payoff matrix (expectancy per 1 risked, before frictions):

WR↓ \ R:R→	1:1	2:1	3:1	1:2	1:3
90%	+0.80	+1.70	+2.60	+0.35	+0.20
75%	+0.50	+1.25	+2.00	+0.25	0.00
60%	+0.20	+0.80	+1.40	+0.10	**−0.20**
50%	0.00	+0.50	+1.00	**−0.25**	**−0.50**
40%	**−0.20**	+0.20	+0.60	**−0.40**	**−0.60**
33%	**−0.34**	0.00	+0.32	**−0.51**	**−0.56**
25%	**−0.50**	**−0.25**	0.00	**−0.63**	**−0.67**

The expert read: 33% WR @ 3:1 = a fine business (+0.32); 60% WR @ 1:3 = a slow bleed (−0.20). The break-even frontier = $p^* = 1/(1+R:R)$: 1:1→50% · 2:1→33.3% · 3:1→25% · 1:2→66.7% · 1:3→75% — every friction moves you further above it. Profitability lives in the PRODUCT — never in either alone.

5.2 RISK-ADJUSTED family — is it a good money machine?

Metric	Definition	Honest role	Trap
Sharpe	mean/std × √periods_yr	comparability of return-per-vol	tail-blind: short-vol strats game it
PSR (Bailey–LdP)	P(true SR > SR*) adj skew/kurt/n	is the Sharpe REAL with fat tails & short samples	Sharpe claims from 60 trades
DSR (Bailey–LdP 2014)	PSR penalized by N TRIALS run	**our certification number** — trials-logged	trial-count amnesia / p-hacking
Sortino	mean / downside-deviation	downside-only quality	upside vol mistaken for risk
Calmar / MAR	CAGR / MaxDD	survivability rating	single-cycle free lunch
MaxDD · time-under-water · recovery	peak-trough depth/duration; gain/MaxDD	governor language: DD states drive de-risk ladders	averaging away the worst state
CVaR / ES	mean loss beyond VaR-tail	the tail-budget number the governor consumes	VaR alone ignoring loss shape
Ulcer Index	RMS of drawdown series	smoothness-of-pain	ignoring suffering time
Information Ratio	active return / tracking error	portfolio skill vs target	mixing benchmark classes

5.3 CONFIDENCE & statistical honesty family

Metric	Definition	Honest role	Trap
Conformal confidence	P(outcome ∈ interval) = 1−α measured OOF	head-contract number; drives sizing/trust	raw model output passed as probability
Reliability diagram / ECE	predicted-prob vs empirical-freq bins	calibration AUDIT per regime	calibration drift post-deploy
Brier score	mean((p−y)²)	one calibration number per head	no regime split
Acc/Prec/Rec/F1	classification quartet	event-cell diagnostics	accuracy alone in rare-event data
PBO (CSCV)	fraction of combos OOS underperforms	design-time overfitting alarm	ignoring until the account does
CPCV	combinatorial purged CV + embargo	multi-path replaces single-path backtest	one lucky path = "proven"
Seed-invariance	same seed ⇒ same weight hash	training determinism gate	silent non-determinism

5.4 EXECUTION & VOLUME family — can we get filled where we said?

Metric	Definition	Honest role	Trap
Implementation Shortfall	realized fills vs arrival price (Perold)	THE live cost number	flattered simulation costs
VWAP/TWAP slippage	deltas vs benchmarks (bp)	participant quality	cross-venue/session mixing
Participation rate	our vol / venue vol	capacity governor input	over-cap → impact explosion
Impact curve	slippage vs size (Almgren-Chriss)	learned per venue; capacity plan	assuming linear at size
Fill ratio / queue-loss	fills ÷ reachable quotes	MM quality; queue reality	backtests ignoring queue loss
Adverse-selection score	mid-move after our fills (signed)	top MM health sign	regime-averaged blindness
ADV capacity	alpha decay vs AUM band	per-cell scaling	static fixed sizing
Turnover drag + frictions	λ·turnover×cost; funding/borrow	decay visibility; real cost lines	gross-only reporting

5.5 PORTFOLIO family — are the parts adding or canceling?

Metric	Definition	Honest role	Trap
Contribution-to-risk	$\rho_i \cdot \sigma_i$ / portfolio σ	who carries the risk	weight ≠ risk
Beta + factor exposures	regression vs market/factors	hidden-bet detection	"multi-cell ⇒ diversified" myth
Regime-conditioned correlation	peer correlation per regime	crowding detection	calm-regime Σ forever
Band compliance	% time in-band	allocator discipline	silent band creep
Cross-agent overlap	position similarity score	systemic single-bet alarm	independence assumption

TOP-10 "what actually matters" (memorize this order)

1. **Expectancy after costs** — the edge question, per regime.
2. **Drawdown states govern capital** — governor ladders act on DD, not returns.
3. **PSR/DSR discipline** — trials logged; deflated acceptance.
4. **Tails (CVaR) in the risk budget** — Sharpe alone is tail-blind.
5. **Execution honesty** — realized shortfall ≤ model claim; else freeze.
6. **Regime persistence** — dwell-time confirmation before reacting.
7. **Capacity curves before scaling** — participation ceilings per venue.
8. **Σ quality before optimizer choice** — shrunk, regime-conditioned covariance.
9. **PIT truth** — no look-ahead; survivorship-certified universe.
10. **Cross-agent crowding** — many cells same direction = ONE bigger bet.

5.7 Gate defaults v1 (tune in first slice)

Gate	Threshold
Type-1 certification	DSR ≥ 0.9 with logged trials; expectancy > control on same folds
MM certification	realized adverse-selection ≤ +10% of model claim; beats static core; pause ≤ 2 events/wk
Portfolio certification	rebalance IS within model ±20%; de-risk ladder fired correctly in replayed crashes
Calibration	ECE < 0.02 per regime bucket pre-deploy
Shadow → live	decision parity ≥ 98%; realized cost ±20% of sim
Latency CI	P99 < 100 µs internal — every merge

6 · RESEARCH-PAPER MAP (each → what it gives us)

Classics are decades-stable; 2025–26 surveys verified live 2026-10-26.

Microstructure & market making

Paper	Gives us
Avellaneda & Stoikov 2008 optimal MM	quote-core math: spread+skew from inventory aversion — our deterministic MM base
Guéant-Lehalle-Fernandez-Tapia 2013	HJB generalization; robust quoting beyond A-S assumptions
Cartea-Jaimungal-Penalva	inventory + adverse-selection dynamics menu

Execution & impact

Paper	Gives us
Almgren-Chriss 2000	impact + TWAP/POV tradeoffs — capacity curves & execution module basis
Perold 1988 Implementation Shortfall	arrival-price cost benchmark family (§5.4)
Bouchaud-Farmer-Lillo LOB laws	empirical impact regularities for model calibration

ML asset pricing

Paper	Gives us
Gu-Kelly-Xiu 2020	supervised edge benchmark standard + feature discipline

Safe / risk-sensitive RL

Paper	Gives us
García-Fernández 2015 safe RL	safety-shield taxonomy ↔ our governor
Achiam et al 2017 CPO	constrained-policy updates
Tamar et al 2016 CVaR RL	tail-first objective training

Evaluation & backtest honesty

Paper	Gives us
Bailey-LdP 2012 PSR	cert stat 1: Sharpe real under fat tails/short n
Bailey-LdP 2014 DSR	cert stat 2: multi-testing discipline → our gate
Bailey et al 2017 PBO/CSCV	design-time overfitting alarm
LdP 2018 Advances in Fin ML	CPCV + meta-labeling (our confidence-sizing layer)

RL in finance (verified current)

Paper	Gives us
arXiv 2512.10913 (Dec 2025) 167-paper review	definitive field map: MM/portfolio/trading RL — course corrections
Wang et al 2025 ESWA survey	componentized view matching our class split
Kolm-Ritter modern perspectives	where RL adds value vs supervised — anti-hype champion framing
Buehler et al 2019 Deep Hedging	options-cell hedging: RL under risk measure; greeks-target action space
BlackRock researchers 2025 RL hedging	production evidence the path matures

Portfolio · regimes · trust

Paper	Gives us
Markowitz 52 · Black-Litterman 92 · Ledoit-Wolf 04	allocator lineage; belief-blend; shrunk Σ default
Kelly 1956	sizing ceiling: full-Kelly banned → fractional cap in governor
Hamilton 89 HMM · Page/Hinkley CUSUM	regime detector + fast change-point design
Lo 2004 Adaptive Markets	edges decay & refill — nightly retrain + evolution rationale
Shafer-Vovk 08 · Angelopoulos-Bates 21 conformal	calibrated-confidence machinery of the Head Contract

7 · Failure-mode defenses (complete)

#	Failure mode	Defense	Enforcement
1	Non-stationarity	regime detectors (CUSUM/HMM) in obs space; rolling windows; DD de-risk	data layer + governor
2	Overfitting	CPCV + purge/embargo; DSR trials-logged; parameter caps	certification algorithm
3	Look-ahead	as_of()-only channel; availability-time features; embargo	physical, data layer
4	Survivorship	membership-at-date universe; certificate else refuse	backtester gate
5	Data leakage	purge/embargo in time + cross-instrument; feature-version isolation	PIT + lineage manifest
6	Regime break (live)	governor capital cut; correct-regime head swap; dwell confirm	deterministic wiring
7	Reward hacking	independent audit-metric set; reward fn versioned; committee reads	committee, not agents
8	Training instability	seed-invariance CI; grad-clip + LR schedule + NaN-watch; min-stability epochs	train gate
9	MM inventory tail	A-S core + certified-bound nets; auto-widen; one-sided pause	quote risk grid
10	Execution-cost drift	realized IS vs sim claim; ±20% → freeze & recalibrate	monitor gate
11	Cross-agent crowding	overlap score; systemic circuit breaker	governor
12	Calibration rot	ECE recheck per regime post-deploy	monitor gate

8 · Build order & honest limits

Milestones

```
M1 Type-1 slice: 1 symbol · trunk+head · replay train
· shadow · HUMAN gate 1
M2 Type-3 lite: static allocator + governor + 1 risk
target
M3 Type-1 variants: derivatives / stat-arb / event (3
engines wired)
M4 Type-2 MM: crypto venue first; fills calibrate
cost model
M5 execution module: when cells scale past starter
bands
each milestone ships: latency CI · seed-invariance CI
· champion race · attribution logs
```

Honest limits (standing) · venue hops bound alpha take-rate; our moats = µs-internal + PIT truth + 981 unique-data connectors + nightly continuous learning + fused 3-engine event context · population edge ≠ per-cell edge — crowding must be governed · RL must beat the GBM race to exist; the supervised control keeps us honest · meta-labeling + conformal confidence = the trust layer, never raw outputs